

Bachelor of Technology — Quantum Science and Engineering

Table 1: Quantum Science and Engineering — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	40
	Discipline Electives	26
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

A dash (—) in the Course Code column means that the source does not yet notify an official catalogue code.

Table 2: B. Tech Quantum Science and Engineering — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EP-301	Engineering Mathematics -2	3-1-0-4	4
2	EP-321	Foundation of Electrodynamics	3-0-0-3	3
3	QS-302 / PH-301	Foundation of Quantum Technology / Quantum Mechanics and Application	3-0-0-3	3
4	PH-302	Statistical Thermodynamics	3-0-0-3	3
5	PH-501	Solid State Physics	3-0-0-3	3
6	QS-407	Introduction to Quantum Sensing	3-0-0-3	3
7	—	Reverse Engineering	0-0-2-1	1
8	CS-212	Design of Algorithm	3-0-2-4	4
9	CS-315 [†]	Artificial Intelligence	3-0-2-4	4
10	QS-304P	Basic Laboratory Course on Quantum Technologies	1-0-4-3	3
11	PH-502	Photonics	3-0-0-3	3
12	QS-501P	Experiments in Quantum Optics	1-0-4-3	3
13	QS-509	Engineering Foundation of Quantum Technologies	3-0-0-3	3
Total				40

[†]CS-305 Artificial Intelligence (3-0-0-3) has been recoded as CS-315 (3-0-2-4) w.e.f. batch 2025; the extra credit is adjusted against Discipline Electives, keeping the total at 160.

Table 3: B. Tech Quantum Science and Engineering — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Quantum Science and Engineering — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE and Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Quantum Science and Engineering — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	EP-301	Engineering Mathematics -2	3-1-0-4	4
3	DC	EP-321	Foundation of Electrodynamics	3-0-0-3	3
4	DC	QS-302 / PH-301	Foundation of Quantum Technology / Quantum Mechanics and Application	3-0-0-3	3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Quantum Science and Engineering — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-201P	Design Practicum	0-0-6-3	3
2	DC	PH-302	Statistical Thermodynamics	3-0-0-3	3
3	DC	PH-501	Solid State Physics	3-0-0-3	3
4	DC	QS-407	Introduction to Quantum Sensing	3-0-0-3	3
5	DC	—	Reverse Engineering	0-0-2-1	1
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Quantum Science and Engineering — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-212	Design of Algorithm	3-0-2-4	4
2	DC	CS-315 [†]	Artificial Intelligence	3-0-2-4	4
3	DC	QS-304P	Basic Laboratory Course on Quantum Technologies	1-0-4-3	3
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Quantum Science and Engineering — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	PH-502	Photonics	3-0-0-3	3
2	DC	QS-501P	Experiments in Quantum Optics	1-0-4-3	3
3	DC	QS-509	Engineering Foundation of Quantum Technologies	3-0-0-3	3
4	ISTP	DP-301P	ISTP or Equivalent	0-0-6-4	4
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Quantum Science and Engineering — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	DP-401P	MTP-1	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Quantum Science and Engineering — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	DP-402P	MTP-2	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 26 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Quantum Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-520	Introduction to Quantum Computing	3
2	CS-671	Deep Learning and Applications	3
3	DS-416	Fundamentals of Machine Learning (CS-669)	3
4	EP-403	Physics of Atoms and Molecules	3
5	ME-353	Electronic Materials and their Applications	3
6	PH-559	Introduction to Quantum Optics	3
7	PH-579	Advanced Quantum Computation and Information	3
8	PH-601	Mesoscopic Physics and Quantum Transport	3

Quantum Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
9	PH-603	Advanced Condensed Matter Physics	3
10	PH-605	Superconductivity	3
11	PH-613	Special Topics in Quantum Mechanics	3
12	QS-301	Survey of Quantum Technology	3
13	QS-406	Introduction to Quantum Communication	3
14	QS-510	Solid State Physics for Quantum Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
15	DP-401P	MTP-1 (compulsory; counted as DE)	4
16	DP-402P	MTP-2 (compulsory; counted as DE)	4